

ActInf Textbook Group ~ Cohort 2 ~ Meeting 3

(Chapter 1, part 2)

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okay well welcome everyone thanks for joining it's september 16th and we are in the third meeting for cohort two we're on our second discussion of chapter one in the textbook so this is our um closing discussion on chapter one which is going to be just our jumping off point into the following chapters we'll go to the questions page where i saw some questions were submitted uh and we can also look at other areas like just the chapter notes more generally but before we go to the questions is there any just remarks that people would like to share on chapter one and then otherwise i'll i'll make one short note about another resource i'll be watching the hand raising too well first maybe i'll just share this other resource and i'll it's already added to the resources section and i'll put it in the nearby it's called pools of insight it's from however um the author is still like around and when i emailed uh them yeah this is the the most recent version and the document is called pools of insight um a previous version i guess was called knowledge hydrant but again this is the most recent version even though it's almost as old as some of us might be um and it's based around alexander's christopher alexander's pattern language which may be familiar to some it's a it's a very unique book and it speaks to archetypes in architecture and design and talks about how there's composition of patterns and makes a language and a grammar for that and that kind of a pattern language approach has been built on in many other settings and this work is a very interesting application of pattern languages to learning groups i think going through it could be interesting for anybody who's just curious about this approach of pattern languages and learning groups and also we're very interested to apply this more intentionally as we continue to iterate on the learning groups format in active inference institute textbook groups for years to come and new textbooks to come and so on just looking at the titles there's a lot of insight like the book is a knowledge hydrant how do we even approach that the group is like a pool of insight how do we approach that you know how are we going to integrate our individual study with group dialogue so not much more to say on this today but if anyone's like excited about that it would

be awesome to hear their um how they see these patterns playing out because a lot of them speak to what we want to cultivate in this group going to the questions page there's still also time for people to be adding in a a brock i put it in the um nearby not the room i'll add it there too and it's also in resources you can just look up you know fools you'll find it people may still add more questions to the page but we have some really rich questions provided already so we'll start with a smaller one in relation to perception action loops the book states the possibilities for control and adaptation are plentiful but very few are useful i'm not sure how useful is being used here useful for evolution survival what determines useful page four what does anyone see or think so i don't know i would just venture to jump up the cliff and guess that maybe the authors mean like full maintenance of the non-equilibrium steady state density or like existence just to guess though all right what's the non-equilibrium steady state it is like the the balance of of all things that we find ourselves in as living creatures so like you know i'm a human i have you know a certain amount of cells tissues and organs i have a temperature that i expect to find my body at and like a range of temperatures that i can survive in i need a certain concentration of oxygen my blood oxygen needs to be maintained at that level so just like all of the things that like i need to have a certain amount of water like like all of the things that contribute to me being able to persist as in my non-equilibrium steady state so it's like you know if you're in equilibrium as a biological organism you are dead life is maintained in this non-equilibrium steady state and i think that we have to be able to maintain that and maybe the authors just need maintenance useful in the maintenance of that great yes so one qualitative way to frame it is with persistence in survival not saying this is the only meaning of utility but persistence and survival are imperatives for things again using thing in the active inference sense as like the particular states the blanket states and the internal for things that are to persist then they have to at least be acting in a survivable way things that are acting as to not persist are not going to be around for long things that are acting as if they are to persist will be repeatedly measured they'll be repeatedly observed by others or they'll be repeatedly observing themselves in the autopoietic case and then blue brought in um a nice term from the actin pontology which is the non-equilibrium steady state and by like mousing over you can see some of the definitions that have been provided and this is all part of the work that people can contribute to if they want to be improving definitions and so on and the first definition is slightly more technical the second definition is slightly less technical and it just says it's a system with dynamics that are unchanging or out of stationarity in some state um that doesn't mean the system is static or unchanging it's like a dynamic balance so we can think about like

the length of a bone isn't a dynamic balance between um being built up and being broken down and so that maintenance of a non-equilibrium steady state is maintained kind of like homeostasis or like body temperature blood sugar etc like blue mentioned in the long run and that may take energy that may be like an exothermic process to maintain there but it also is maintaining a stasis in some other respect so again few are useful what is this first part why are the possibilities for control and adaptation plentiful and why does that present an issue ali uh i was just gonna mention the three phases of bayesian mechanics i mean mode matching dynamics mode tracking dynamics and path tracking dynamics from which active inference dynamics falls in the category of the path tracking dynamics but unlike the first two types i mean mode matching dynamics and mode tracking dynamics which requires density i mean which requires ness assumption uh the path tracking dynamics yeah that's it uh is present is described as a known as assumption dynamics so if we extend the active inference framework to include not only i mean nest preserving dynamic to all the other kinds of agents which uh just do this kind of path tracking dynamic then uh i guess we would not need uh any kind of nest preserving mechanism to do these kind of action perception the perception action loops very interesting um this is bringing in another resource which is an excellent contemporary resource which is some of the recent work of ramstead at all on on bayesian mechanics and it's it's unpacked in guest stream 23.1 so that that's one very nice resource um and coming back to what blue set about non-equilibrium steady states we can think about one case as being like there's a fixed point and we want just simple convergence to a fixed point and that might be useful like if utility in a situation is defined as we want the plate to be balanced on the stick and we want it the plate to be as tall as possible that's like kind of a classical control theory setting then we might want to have that one fixed mode and that's our utility it's not some sort of like absolute theological utility it's just saying like within this domain or aspect or dimension utility is going to be defined by reducing the divergence to some fixed mode alternatively there could be a dynamic mode following like um there's going to be some cursor on your screen and you're just going to be seeking to follow it and in that case utility is going to be defined by the divergence minimization to a moving mode and then the last case is this path matching which does introduce some um more powerful in general mathematics like um the gauge formalism and in that case the path distances are being divergence minimized it brings in a super important topic that that um next i i um we'll turn to brock to give a thought on which is this idea of utility and pragmatic value as divergence minimization we're gonna come back to that but brock feel free to add what you would like about this if you want it's

page four so it might be there's a lot more like as we go through this that will kind of start helping to connect dots but um in the kind of biological uh homeostasis can like example the blues there are just a lot of things that you could do that would not be like you're saying in that context useful for the organism um and there are things that you could do that are useful maybe in a moment but they're not kind of meta useful like or like creating more usefulness or something like this for the organism right but there are are things that that an organism might do that like um just kind of put it in a place where all the things it could do are just significantly more likely to be and so one of those things that it might do is like go find forage for like a place where food is plentiful and then if it gets hungry and it just starts to go look then necessarily it will kind of start finding food more readily than if it hadn't gone to that patch or you know humans we build houses or some sort of structure that protects us from the elements and then you know um if we oversleep or whatever like a wild animal doesn't get us and neither does the cold or the maybe in the same way like um so that's the language that the book is going to use around that is about free energy reduction like free energy which is just to say in some really coarse grain not completely correct way that like it's just taking a lot of choices off of the table a lot of actions that um would not help would not cause the thing to persist or find more usefulness or and and the other two words here that i'm using a pragmatic epistemic value is the epistemic value is kind of like going to find the like foraging for the food and then the the pragmatic value is like like actually eating getting the food or something like that um not exactly but very rough analogous sort of i don't know if that's helpful but that's awesome thank you yeah many pieces here so first you mentioned like the state space of possible moves which is what is being suggested here with the first part of the sentence the are plentiful so we can think about those in terms of affordances which are relational capacities for action and the set of possible moves or affordances might be large however only like a vanishingly small subset of them might be useful and then part of the challenge that's pointed to is that um there's a recursive nature of the cycle so it's not like we can just say well you know pawn to e4 is just simply a good move because there's probably situations where that like wins you the game and there's probably situations where that loses the game within the utility defined by like a game there's already immense challenges in exploring these like branching trees of outcomes and just evaluating them under any metric period how can we evaluate perception and action planning the idea that's going to come into play is action and planning as divergence this can be contrasted with the notion of perception cognition action planning etc as reward maximization reward maximization is equivalent to pragmatic value

utility optimization is pragmatic value u_m and we're going to see this reflected soon some of the equations to come in the next chapters but for chapter one we're still addressing this like u_m qualitatively pragmatic value is like the task you know time on task u_m if if the pragmatism in the mountain climbing example is climbing the mountain then pragmatic value is like every inch you take up the mountain epistemic value would be reducing your uncertainty about how to achieve pragmatic value and so in that way the free energy functional which encompasses both pragmatic and epistemic value allows a more adept navigation of explore and exploit because under a reward maximization framework it's very easy to like you know turn the speaker up to 11 set the phaser to stun and pursue local pragmatic value all of these second order heuristics need to come into play to reintroduce like novelty bonuses curiosity as a second level on to this u_m reward first pragmatic value first foundation and in contrast the imperatives for action and perception in active inference are going to be built around not reward maximization but free energy minimization or relative free energy reduction as mentioned and that's going to be composed of there's going to be multiple ways to break that down what the free energy functional is it is going to often come down to u_m a pragmatic element in an epistemic element u_m the pragmatic element is going to look like again we're just beginning to come into these sections the pragmatic value is going to look like a reduction of divergence between preferences and observations so instead of saying well the most rewarding body temperature for me is 37 and i'm maximizing reward by staying at that's a reward maximization perspective on temperature homeostasis a divergence minimization perspective surprise minimization perspective on body temperature homeostasis is i expect and prefer to find myself at 37 and i'm going to undertake policy to reduce my surprise about finding myself at 37. so when i'm at 37 i'll be minimally surprised and as i get further i will be more and more surprised and then i will take pragmatic action to climb that hill as well as epistemic actions to find out how to climb that hill but we're going to be getting back to 37 by reducing the divergence between preferences and outcomes rather than getting to 37 by like winning and trying to maximize reward which is the only fundamental u_m ontology for reward based learning so that's how many people have in that basal relationship between pragmatic and epistemic value and active inference that's where many people have found space to explore like curiosity and novelty and a lot of other things will open up once we first start to explore how pragmatic and epistemic value are like connected and as coequal in the imperatives for active inference agents and then there'll be even more fun things to come like again when we realize that pragmatic value isn't maximizing rewards simply it's reducing the

divergence relative to expectations so it's like again not um we're playing chess and we're trying to maximize our points and minimize their points but we have an optimistic trajectory or set of expectations and then we're going to reduce the divergence such that that is realized so what determines useful as with many other questions there's like a limited answer that's so limited that it may not feel like an answer which is like again if we're balancing a plate on a stick we may define utility as keeping the plate up if we were climbing a mountain we may define utility in that limited dimension as getting to the top of the mountain that may feel like too narrow because we might think well like but the person balancing the plate i mean they have to have a salary and they have to breathe and have blood sugar and stuff like that the limited answer is like sort of just staying within the bounds of our model and the dimensions that we're considering and then part of this iterative modeling process which especially starts to be introduced in the second half of the book but also is reflected by people's experience other modeling areas including qualitative modeling areas like so then it's okay so now we have the plate balancer and their blood sugar and then are we still missing something relative to the real world are we as modelers still being surprised by what happens now we need to consider something else that's like the limited answer which is utility is situational and dimensional in a narrow sense and then in perhaps a slightly more like miso or macro sense um or at least like um a cousin or a proxy of utility is survival and persistence because without survival and persistence you're not going to be around to even have that debate about what's so it is kind of like a prerequisite or an enabling feature of any dimensional or like sort of one aspect you know if you want to be in the game [Music] to realize any specific kind of utility being in the game is like that meta brock pointed towards with like different um behaviors and phenomena ranging from epistemic foraging to um niche construction and like building a house any other thoughts on utility what determines useful this looks like a great long question so i'm going to just bring it down so we can kind of break it into sections can anyone describe the neat and scruffy distinction it's on page five ollie yes it's again related to the high road on the low road i mean uh they've described the high road as the needs perspective or the needs approach and the low road as the scrappy one uh probably because high road begins with some some established postulates and let's say without going into a bottom-up approach and beginning with the fundamental principles and then developing uh the propositions uh from there but beginning with established propositions and then somehow chopping it off or somehow going into a much more granular level in terms of its fundamental constituents thank you excellent i i find this paragraph to be a very nice point which is we can think of two

extreme ends of a um kind of like research program or epistemic ecosystem one would be every single neural process you know acetylcholine synapses neuro hormones being circulated in this part every single physiological piece could have its own theory i mean they could have their own department they could have their own university um and same for every cognitive mechanism we could have a separate bespoke model with no interoperability at all there's no reason why it has to be between um attention cognition perception action social interaction communication anticipation planning we could have bespoke theories and somebody might suggest that that is like preferable in their eye whether for utility or for aesthetic reasons in their words that would lead to a proliferation of theories in fields such as the above with little hope for their unification another perspective is that despite their diverse manifestations the central aspects of behavior cognition adaptation in living organisms are amenable to coherent explanation from first principles so in other words could we start with first principles can explain or explore what those are but can we start with first principles and then see different biological processes and different cognitive mechanisms as being under a common umbrella or under a common ontology or addressable by a shared model then to come back to the question um because this is a nice key point of this uh chapter it's uh alluding to roger shank introducing this neat versus scruffy distinction and then as ali suggested this does map to the high road and the low road which um are mapped out in figure 1.2 the neats are like the lumpers they're finding unity heterogeneity and then the scruffies are like embracing the heterogeneity by focusing on details that demand dedicated explanations i mean how can we have a model of all synapses when the acetylcholine synapses are so different than the dopamine synapses and then we'll need a third model or a third paper to describe when there's both acetylcholine and dopamine at the same synapse so that's like the scruffy bottom up embracing the heterogeneity of nature and then the neat is coming from the top down or from the high road and finding unification across that diversification brock um sorry i just read this whole question and i don't want to kind of blow past um getting into the details of each but i want to try to maybe from where you started there kind of thread uh the needle there too to the end of what's kind of being asked here is that well he's he's in the context of these you know uh bottom up kind of less more explanatory but less meaningful and the top down kind of more meaningful but maybe in some cases less explanatory or something like this like how do we look at active inference or where does it sit in relation to all these other um theories and models of mind body cognition etc and specifically in in fristen's um essay uh the desert landscape here um he's got a quote about fep kind of subsuming all other theories as not some

prediction but just that as they kind of take one at a time here they have kind of just happened to be commensurate and be describable in some way here by fep and so is that kind of like meaningless i lost you rock i'm not sure if that was me hmm you well sorry about this not sure what is literally happening with my internet loading okay yep uh kind of a source of the confusion for most people about what exactly what does active inference exactly explain or uh and what's the nature of the explanation there can you um copy paste that that's a good response and i think i've appended uh that response to the pdf that i've uploaded course's page i was just i was just going to say that that you know the desert landscape being left there is is kind of like a new niche that looks maybe barren today but if we continue maybe it's not so barren maybe there are adjacent things that are that emerge from that niche being constructed that cultural lens being used that we look out and see something different that we weren't expecting sorry about my strange absence bronwyn i think okay yeah brock yeah are you daniel uh john and you are talking learners yeah i can't hear i'm gonna read life in a more simple frame from my my perspective which is not as advanced as anyone here in the discussion but it's but it's more from from exactly uh the descriptions you've been or what you've been talking about has made it a bit more clear in a way i'm not sure about the high road and the low road sort of bottom up bottom down thing but you know there's a lot of theories about everything but what was what came through then to me was that with fep and active inference it's a bit um like it hasn't been fully explored like it's it's um that it's something where we're actually investigating ourselves so we don't actually know what we're going to come up with a bit like the sort of elephant with a whole lot of people blind people trying to work out what the elephant is and and not actually getting a grasp of the whole thing and um which makes it something that's going to that still as someone said still evolving still actually developing into whatever it will be eventually yeah which is which is helpful for my for my question in terms of if it's um if it's going to consume every other theory makes it a little bit too grand and a little bit too much i think i don't know if i'm making sense but um it's quite a complex um yeah i i think that that is that's that is a very common response and it's still i mean when you say you're not as advanced as i mean uh we're just using words we've learned over the last two months maybe some of us um speaking personally um so you know it's again there's this question like oh if it describes everything or you know and it's unfalsifiable then what what does that mean you know it's that's how is that helpful or what is the use there i think what what you keyed in on this sort of evolving that we don't know quite yet and how ali um called attention to that response from carl of a kind of cultural lenses being developed like

that that is kind of like niche construction that is that is us figuring out what is the extent of this niche where does fap apply where does active inference apply like really well yeah and then where does daniel always uses linear regression because it's so easy to grasp at like there's many cases where you know maybe active inference maybe some magical process is happening but wow linear regression works really well you know so you know active inference applies but it's a lot of work and linear regression works really good right um so there are certainly i would expect you know boundaries to where the pragmatic use like the usefulness of it is not it doesn't um continue there's some drop-off redu yeah so we don't know what that is precisely yet i certainly don't i don't know if somebody else does but um it's it's a question that's being explored and i don't yeah i think we just will figure that out over time through exploring and finding where that desert you know ends kind of or slowly turns back into an oasis for us to forage in nice thank you for this um discussion i do love the linear regression i mean imagine if i said really numbers you think you're going to be able to describe apples and oranges with numbers or height and weight i mean could two things be more different than time and height and weight you really think a linear regression is going to cut through that um so sometimes modeling and statistics overall can feel like they're encompassing a lot and then especially when there's so much philosophy and reflection and like relational thinking in the mix it truly can converge on what feels like a theory of everything one great video is um from a channel called theories of everything by kurt jeimungul and um it's with carl first and there's several with carl and so that is tackled head on and i don't exactly recall whether he goes this direction but the the the um quintessentially fristonian twist is yeah we are talking about everything everything that for you is in this sort of like process object dialectic it is realizing a non-equilibrium steady otherwise you wouldn't call it a thing so whatever it is that separates that thing from what is not that thing is like a blanket and if you're gonna say it's a fuzzy blanket well some blanket index and there's a space for that discussion as well um but when we're talking about things as they are and or as we model them we can put them into a framework of particular things and that's the free energy principle for a particular physics also known as the bayesian mechanics so it it does take a slightly different approach and then um what will active inference converge upon or move between these are definitely open and developing questions and and also i i like what like about the desert and the oasis because the desert is alive as well it's it's a reference to this um to quine's desert landscape and some of that work which has been referenced very um interesting and then there's also like a sort of interpersonal approach slash like rhetoric like is

this presented as are you presenting actin from the high road well all things need to resist dissipation so that's why we're using non-equilibrium steady states to describe how things exist and those kinds of systems have to be doing something that either is active inference or that we can describe as active inference and starting with an imperative of persistence coming from the low road and thinking about instances and how we can model what they're doing using bayesian methods and many times people will find resonance with either or both of those roads there's a huge amount of ontology mapping availabilities for people to connect different theories to each other and also with ali and jakob and ander well next month with the bayesian mechanics we'll we'll tackle this beijing mechanics question head on so it's just i mean year after year and month after month there's a lot to learn and understand here and sometimes the the philosophers um outrun and anticipate where things may go and then sometimes the writing and the genre can feel like something has already been achieved like well there's already been a synthesis of an activism an act of inference or it can feel like something that is being um doubted is like a conclusive statement that it cannot be addressed markov blankets cannot address physical interfaces sometimes lifted from the modeling there can be writing and conversation that might seem like it is like more or less confident about the status or future or past of this area which is like what motivates this kind of like reading between the lines and between the citations and also a deflationary modeling approach where like um we don't know you know it's still it's still in evolution as to to what to what it does but it sounds as if it'll evolve into explaining some things but not all things i suppose that's where i've got to at the moment at this point yeah and just you know the overriding i'm not sure i wouldn't worry about it overriding um theories or you know things that kind of already are useful um right it's you know relativity doesn't make you know newton's uh laws irrelevant right where that boundary is specifically is a difficult question right now but i think the vast majority of theory models of these cognition are not that are you know historically been useful are not being thrown out they're just being we're trying to connect them right so i would also just add to that um there's like the instrumentalist debate versus the realest debate and i think that um you know the act of inference is not necessarily to be taken as this is not the way that it is more that it's a tool that we can use in our in our toolbox and so i don't know that it invalidates any other theory um but it is um just another another tool in the toolbox right another thing that we can use to model things and definitely like there are some things that active inference or at least as i see it it doesn't explain yet like like self-sacrificing behavior right like like it doesn't necessarily explain that yet but maybe if you look across different temporal skills that that's

like possible or like in yeah like negative harm and harmful like self-harm so active inference i don't think um maybe we'll ever successfully model that but also like one more thing i just want to mention um i don't think anyone here or i don't want to speak for everyone but i don't think anyone in the classroom today feels like they're at the top of that learning curve so um like i've been studying active inference for like years already and i still feel like maybe i'm in the long tail end of the trail on the way up the mountain so um just just just know that it's definitely a process yeah i think i've realized that i mean i've been reading about it for a few years but um and in relationship to to to what i do um as a i'm an alexander technique teacher so i'm looking at it from that perspective and just seeing how it fits in um and so it comes up against other uh motor control theories and you know i mean to say something really simple is uh the idea that we that we don't have motor commands we have predictions um just the difference between that and if i was going to put forward active inference to my colleagues then then how do i substantiate that or how do i explain that this is just a question in my mind i suppose do i say that you know the motor command thing's out the window and it's all uh prediction or do i just say this is a possibility this is one idea yeah so that's a great point point is that daniel any funny i can i can hear you though so it's on the recording um anyone else have anything asked comments or nice it's it's great yes go ahead so i exchanged some emails with the thomas parr because i went through the book and you know i noticed some things that may have been some typos or some or ask him some some questions and so i i compiled the document on overleaf with these points and questions and it was a very kind to answer and to provide answers and about also some clarification of some i'm i sent him an email and asked him if he was okay to share it with the group to share the document with the group and he hasn't answered yet but i'm pretty sure he will consent and then you know thank you so in this errata section here's where we've um have um a handful of uh typos and ambiguities and i'll make a subpage you'll have full control so just anything you want to do in this subpage just do it okay um all right well great time sorry about my strange internet just disrupts beyond belief we did it i'll uh upload this soon and uh enjoy reading through chapter two please add any question the most basic and the most confirmatory to the most speculative every question will be really amazing to see for chapter two so thanks everyone and see you soon